Wiki :

$$\begin{pmatrix} 1 & 2 & 3 \\ \sigma(1) & \sigma(2) & \sigma(3) \end{pmatrix}$$

rearrangement of 1 2 3

$$\begin{pmatrix} 1 & 2 & 3 \\ \sigma_a(1) & \sigma_a(2) & \sigma_a(3) \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ \sigma_b(1) & \sigma_b(2) & \sigma_b(3) \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ \sigma_{ab}(1) & \sigma_{ab}(2) & \sigma_{ab}(3) \end{pmatrix}$$

$$\sigma_{ab}(1) = \sigma_a(\sigma_b(1))$$

$$\therefore \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$1 \to 3 \to 2$
$2 \to 1 \to 1$
$3 \to 2 \to 3$

保持 box 凸 凸 凸 位置不变，沿着乘法顺序，保持出发的盒子编号，更换不同编号的粒子（successively）。

α β γ : wave function

展开 $\begin{pmatrix} 1 & 2 & 3 \\ \sigma(1) & \sigma(2) & \sigma(3) \end{pmatrix}$ $\alpha(1)\ \beta(2)\ \gamma(3)$

$= \alpha(\sigma(1))\ \beta(\sigma(2))\ \gamma(\sigma(3))$

交换粒子的编号，将求和 α, β, γ

顺序排开，使用连乘将形式方便

box : wave function

object : particle

$\begin{pmatrix} 1 & 2 & 3 & \cdots \\ a_1 & a_2 & a_3 & \cdots \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 & \cdots \\ a_3 & a_1 & a_2 & \cdots \end{pmatrix}$

重要的是匹配关系

即 $S_3 \approx D_3$ 中 $C = FB$

e.g. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$

量子力学：全同粒子

（W. Greiner）

中文译本：《量子力学：对称性》

北京大学出版社

$$\varepsilon_{ij} = \varepsilon_{ji} \qquad \varepsilon_{ij}^2 = 1$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \alpha(1)\beta(2)\gamma(3)\delta(4) = \alpha(2)\beta(3)\gamma(4)\delta(1)$$

将人为编号为 2 的粒子置于 α 态

3	β
4	γ
1	δ

$$\alpha(2)\beta(3)\gamma(4)\delta(1) \equiv \delta\alpha\beta\gamma$$

（默认 1 2 3 4 的粒子编序）

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \alpha\beta\gamma\delta = \delta\alpha$$

$$\hat{P}_{14}\hat{P}_{13}\hat{P}_{12} = \begin{pmatrix}1&2&3&4\\4&2&3&1\end{pmatrix}\begin{pmatrix}1&2&3&4\\3&2&1&4\end{pmatrix}\begin{pmatrix}1&2&3&4\\2&1&3&4\end{pmatrix}$$

$$= \begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix}$$

start with right-most permutation

用 $\psi(1,2,3,4) \leftarrow \alpha(1)\beta(2)\gamma(3)\delta(4)$

则 $\psi(2,3,4,1) \Leftrightarrow \alpha(2)\beta(3)\gamma(4)\delta(1)$

↓

$\delta\alpha\beta\gamma$

S_2 : $\psi(1,2)$

flavor SU(2) 强相互作用不变

1 (position, spin, isospin) $\vec{r}_1, s_1, I_1$

2 $\vec{r}_2, s_2, I_2$

一般情况，$\psi(1,2)$ 单无对称性

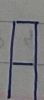
$\Rightarrow \hat{P}_{12}\psi(1,2) = ?$

$\psi_S = \psi(1,2) + \psi(2,1)$

$\psi_a = \psi(1,2) - \psi(2,1)$

问：ψ_a, ψ_S 是否为换位算符的本征态？ $\begin{cases}\hat{P}_{12}\psi_S = \psi_S \\ \hat{P}_{12}\psi_a = -\psi_a\end{cases}$

非简并多重态，单态

例子：$\psi_S =$ $\psi_a =$ 

一个粒子对应一个格子

$$S_{ij} = (I + P_{ij})$$

$$S_{ij}^2 = (I + P_{ij})(I + P_{ij}) = 2 S_{ij}$$

$$A_{ij} = (I - P_{ij})$$

$$A_{ij}^2 = (I - P_{ij})(I - P_{ij}) = 2 A_{ij}$$

S_{ij} , A_{ij} 是投影算符

作用 S_{ij}

已经对称化的 state $|\psi_s\rangle$ 再次对称化 还是原来的 state $|\psi_s\rangle$

已经反对称化的 state $|\psi_a\rangle$ 再次反对称化 还是原来的 $|\psi_a\rangle$

作用 A_{ij}

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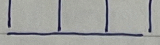
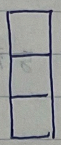
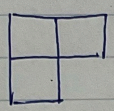
S_1 : □ → one particle, trivial case

S_3 : 三粒子情形 (三个杨图)

ψ_S =  (fully symmetric state)

ψ_a =  (fully asymmetric state)

ψ_m =  (mixed symmetric state)

↓ 两个都在 一好的粒子 交换对称

(一列) 两个粒子 反对称

Identity operator ($\mathcal{I}$) 书中 ↓ 改用 I

对称化算符

$\hat{S}_{ij} = I + \hat{P}_{ij}$, $\hat{A}_{ij} = I - \hat{P}_{ij}$ (对比 ψ_a, ψ_S 可知)

$\hat{S}_{ij}\hat{A}_{ij} = I + \hat{P}_{ij} - \hat{P}_{ij} - \hat{P}_{ij}^2 = 0$ (练习)

$\hat{A}_{ij}\hat{S}_{ij} = 0$

以 $\psi(1,2,3)$ 开始, 混合对称 ψ_m ($m = 1,2,3,4$)

$\psi_1 = \hat{A}_{13}\hat{S}_{12}\psi(1,2,3)$ $\xrightarrow{\hat{P}_{12}}$ $\psi_2 = \hat{A}_{23}\hat{S}_{12}\psi(1,2,3)$

$\psi_3 = \hat{A}_{23}\hat{S}_{13}\psi(1,2,3)$ $\xrightarrow{\hat{P}_{13}}$ $\psi_4 = \hat{A}_{12}\hat{S}_{13}\psi(1,2,3)$

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另两个基函数 ψ_2 ψ_3 (1)

ψ_1 ψ_4 (2)

2,3 交换反对称，与 (1) 不一样（1,2 反对称）

ψ_{12}

(1),(2) 通过 $\psi_{12}, \psi_{13}, \psi_{23}$ 可相互转换

$\psi_{12} \leftrightarrow$

$\leftrightarrow \psi_{31}$

两个不变子空间（两个双重态）

$$\lambda \psi_{\square} + \mu \psi_{\square} \quad (3)$$

$$\lambda' \psi_{\square} + \mu' \psi_{\square} \quad (4)$$

two standard forms cannot be transformed into each other by permutations of the group S_3

(3) 式代表的波函数中对称1,2的任意置换给出的波函数是两个基矢的线性组合

(4) 证一样

例 $P\left(\begin{smallmatrix}1&2&3\\2&3&1\end{smallmatrix}\right)\left(\lambda \psi_{\square} + \mu \psi_{\square}\right)$

$\psi_{\square}$ 与 $\psi_{\square}$ 不能通过任何对称元连接

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$= -(\lambda+n)$ [1 2 / 3] $+ \lambda$ [2 1 / 3]

两个点子空间可以通过群元连接 (×) 不行

但其表示矩阵一致？ 不要混 S_3 本身的表示。

特征标：按！ 和三个全同粒子在 S_3 置换下的对称性讨论（关于波函数）

Burnside 定理

一个不恒等的两维不可约表示。

determine the basis functions of the permutation group S_3 in terms of the single particle wave functions $\alpha(i)\ \beta(j)\ \gamma(k)$

$\psi_S =$ [1 2 3] $= \hat{S}_{123}\ \alpha\beta\gamma$

$\hat{S}_{123} = I + \hat{P}_{12} + \hat{P}_{13} + \hat{P}_{23} + \hat{P}_{13}\hat{P}_{12} + \hat{P}_{12}\hat{P}_{13}$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

normalizing ψ_S we get

$\psi_S = \frac{1}{\sqrt{6}}(\alpha\beta\gamma + \beta\alpha\gamma + \alpha\gamma\beta + \gamma\alpha\beta + \gamma\beta\alpha + \beta\gamma\alpha)$

Similarly for the antisymmetric wavefunctions:

$\psi_a =$ [1/2/3] $= \hat{A}_{123}\ \alpha\beta\gamma$

$\hat{A}_{123} = I - \hat{P}_{12} - \hat{P}_{13} - \hat{P}_{23} + \hat{P}_{13}\hat{P}_{12} + \hat{P}_{12}\hat{P}_{13}$

$\psi_a = \frac{1}{\sqrt{6}}[\alpha\beta\gamma - \beta\alpha\gamma - \gamma\beta\alpha - \alpha\gamma\beta + \gamma\alpha\beta + \beta\gamma\alpha)$

refer to P318-320 Quantum mechanics symmetries 2nd Hassin W. Greiner B. Müller

No.

Date

Schmidt orthonormalization procedure

$\psi_1 = \begin{array}{|c|c|}\hline 1&2\\\hline 3\\\cline{1-1}\end{array} = \hat{P}_{13}\hat{S}_{12}\,\alpha\beta\gamma = \alpha\beta\gamma + \beta\alpha\gamma - \gamma\beta\alpha - \gamma\alpha\beta$

$\psi_2 = \begin{array}{|c|c|}\hline 2&1\\\hline 3\\\cline{1-1}\end{array} = \hat{P}_{23}\hat{S}_{12}\,\alpha\beta\gamma = \alpha\beta\gamma + \beta\alpha\gamma - \alpha\gamma\beta - \beta\gamma\alpha$

$\psi_3 = \begin{array}{|c|c|}\hline 1&3\\\hline 2\\\cline{1-1}\end{array} = \hat{P}_{12}\hat{S}_{13}\,\alpha\beta\gamma = \alpha\beta\gamma + \gamma\beta\alpha - \beta\alpha\gamma - \beta\gamma\alpha$

$\psi_4 = \begin{array}{|c|c|}\hline 3&1\\\hline 2\\\cline{1-1}\end{array} = \hat{P}_{23}\hat{S}_{13}\,\alpha\beta\gamma = \alpha\beta\gamma + \gamma\beta\alpha - \alpha\gamma\beta - \gamma\alpha\beta$

they are not orthogonal

$\psi_2' = \psi_2 + a\psi_1$ 要求 $\langle\psi_1|\psi_2'\rangle = 0 \Rightarrow a = -\frac{1}{2}$

$\frac{1}{\sqrt{2}}\psi_1$, $\frac{1}{\sqrt{3}}\psi_2'$ are an orthonormal basis of a two-dimensional irreducible representation of S_3, i.e. they form a doublet.

we proceed to construct a basis function from ψ_3, ψ_1, ψ_2'

$\langle\psi_3'|\psi_1\rangle = 0$, $\langle\psi_3'|\psi_2'\rangle = 0$

$\psi_3' = \psi_3 + \frac{1}{4}\psi_1 - \frac{1}{2}\psi_2'$

$= \alpha\beta\gamma - \beta\alpha\gamma + \frac{1}{2}(\gamma\beta\alpha + \alpha\gamma\beta - \gamma\alpha\beta - \beta\gamma\alpha)$

$\psi_4' = \psi_4 - \frac{1}{4}(\psi_1 + 2\psi_2' + 2\psi_3') = \frac{3}{4}(\beta\gamma\alpha + \gamma\beta\alpha - \alpha\gamma\beta - \gamma\alpha\beta)$

here S_3 transforms the functions ψ_3' and ψ_4' into linear combinations of the same functions again.

$\frac{1}{\sqrt{3}}\psi_3'$, $\frac{2}{3}\psi_4'$

an orthonormalized basis of a second doublet

In terms of group theory, a set of degenerate states is called a multiplet.

S_3	N_C		1	$\bar{1}$	2
	1	I	1	1	2
Z_3	2	$(123)(132)$	1	1	-1
Z_2	3	$(12), (23), (31)$	1	-1	0

# of elements in a class

even permutation : 1

odd ~ : -1

$(123) = (12)(23)$

$(132) = (13)(32)$

$(123)(123) = (132)$

$(123)(132) = I$

cyclic group : $A_3 \cong Z_3$

二维表示 (D, F 是对角矩阵) 的构建 :

From the character table to the representation matrices

Let's go to a basis in which (123) and (132) are diagonal.

then $\begin{pmatrix} w_1 & \\ & w_2 \end{pmatrix}$ $\begin{pmatrix} w_1^2 & \\ & w_2^2 \end{pmatrix}$

$$w_1^3 = w_2^3 = 1$$

unitary : $\begin{pmatrix} w_1 & \\ & w_2 \end{pmatrix} \begin{pmatrix} w_1^* & \\ & w_2^* \end{pmatrix} = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$

$$\downarrow \quad w_1 = w_2^*$$

then

(123) : $\begin{pmatrix} w & \\ & w^* \end{pmatrix}$ (132): $\begin{pmatrix} w^* & \\ & w \end{pmatrix}$

$w + w^* = -1$ (from character table)

$$e^{i\frac{4\pi}{3}} + e^{-i\frac{4\pi}{3}} = -1$$

what about (12), (23), (31) ?

pauli matrices : $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

traceless matrix.

character = 0

$tr(\sigma_{x,y,z}) = 0$

$\sigma^2 = 1$ (2度简并)

结合后面讨论的两种不变量.

$$\begin{cases} g_1 + \omega g_2 + \omega^2 g_3 \sim \varphi_1 \\ g_1 + \omega^2 g_2 + \omega g_3 \sim \varphi_2 \end{cases}$$

$A_3 \varphi_1 \sim \varphi_1$, $A_3 \varphi_2 \sim \varphi_2$

$\hat{P}_{12} A_3 \varphi_1 \sim \varphi_2$, $\hat{P}_{12} A_3 \varphi_2 \sim \varphi_1$

$\{ A_3, \hat{P}_{12} A_3 \} \cong Z_2$

S_3 / A_3

Table 17.3. Extended character table for permutation group $P(3)$

	χ(E)	χ(A,B,C)	χ(D,F)	
$P(3)$	(1^3)	$3(2,1)$	$2(3)$	
Γ_1^s	1	1	1	
Γ_1^a	1	−1	1	
Γ_2	2	0	−1	
$\Gamma_{\text{perm.}}(\psi_1\psi_1\psi_1)$	1	1	1	$\Rightarrow \Gamma_1^s$
$\Gamma_{\text{perm.}}(\psi_1\psi_1\psi_2)$	3	1	0	$\Rightarrow \Gamma_1^s + \Gamma_2$
$\Gamma_{\text{perm.}}(\psi_1\psi_2\psi_3)$	6	0	0	$\Rightarrow \Gamma_1^s + \Gamma_1^a + 2\Gamma_2$

odd perm

even perm

好 basis　　重排　　正规表示分解

A, B, C : odd permutation

2维表示。

一元三次方程的三个根的排不变性（任意交换两个不变）

以$|\Gamma_2\alpha\rangle = a + \omega b + \omega^2 c$，$|\Gamma_2\beta\rangle = a + \omega^2 b + \omega c$为基，计算表示矩阵

对角元为0

特征标为0

$$E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & \omega \\ \omega^2 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & \omega^2 \\ \omega & 0 \end{pmatrix} \quad D = \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix} \quad F = \begin{pmatrix} \omega^2 & 0 \\ 0 & \omega \end{pmatrix}$$

$\chi(E) = 2$

$\chi(D) = \chi(F) = \omega + \omega^2 = -1$

$\chi(A) = \chi(B) = \chi(C) = 0$

S_n has an n-dimensional representation, its defining or fundamental representation.

for S_3, dim-3 representation is reducible

$3^2 > 3!$

treat 1 → $\begin{pmatrix}1\\0\\0\end{pmatrix}$, 2 → $\begin{pmatrix}0\\1\\0\end{pmatrix}$, 3 → $\begin{pmatrix}0\\0\\1\end{pmatrix}$

(12) : $\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}$ $\quad \chi_{(12)} = \chi_{(123)} = \chi_{(13)} = 1$

$(12)\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}0\\1\\0\end{pmatrix}$ $\quad (12)\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\end{pmatrix}$

(123) : $\begin{pmatrix}0&0&1\\1&0&0\\0&1&0\end{pmatrix}$ $\quad \chi_{(123)} = \chi_{(132)} = 0$

representation 3 :

3	I
0	z_3
1	z_2

$= 2 \oplus 1$ (特征标表的 1, 2 表示)

$= \begin{array}{|c|}\hline 2\\\hline -1\\\hline 0\\\hline\end{array} + \begin{array}{|c|}\hline 1\\\hline 1\\\hline 1\\\hline\end{array}$

Another way: systematic

P234 李新征

杨算符是杨盘T的算符 $\hat{P}(T)$ 与 $\hat{Q}(T)$ 的乘积，

形式为 $\delta_q = 1$ for even perm, -1 for odd

$$\hat{C}(T) = \hat{P}(T)\hat{Q}(T) = \sum_{\hat{P}\in R(T)} \sum_{\hat{Q}\in C(T)} \delta_q \hat{P}\hat{Q}$$

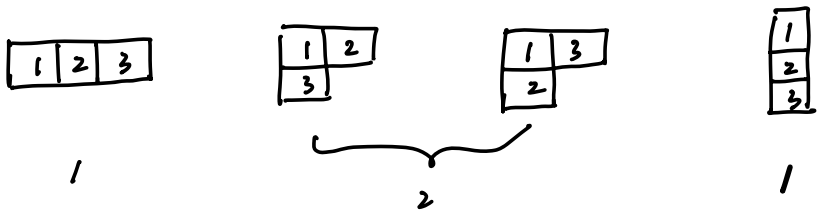
$$= \left(\sum_{\hat{P}\in R(T)} \hat{P}\right)\left(\sum_{\hat{Q}\in C(T)} \delta_q \hat{Q}\right)$$

$\hat{C}(T)$ 是群空间 R_{S_n} 中的一个矢量，一个杨盘对应一个杨算符。

$R(T)$：杨盘所涉及的行置换（同行 intra 置换）（行内，非不同行）

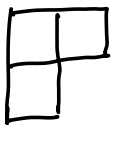
$C(T)$：列置换（同列

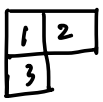
三阶循环群杨准盘



共4个杨准盘 = dim of irreducible representation

求 [2,1] 的不可约表示. (2-dim)


杨图 [2,1]

从标准盘 

出发，其标算符为：

$$R(T) = \{(1),\ (1,2)\}$$

$$C(T) = \{(1),\ (1,3)\}$$

$$\hat{e}(T) = \{(1) + (1,2)\}\{(1) - (1,3)\}$$

$$= (1) + (1,2) - (1,3) - (1,3,2)$$

不可约表示空间为 $R_S \hat{e}(T)$，维数是 2。因此，实际定义所基，求表示矩阵。

$(1)\ \hat{e}(T) = \hat{e}(T)$

$(1,2)\ \hat{e}(T) = \hat{e}(T)$

$(1,3)\ \hat{e}(T) = (1,3) + (1,2,3) - (1) - (2,3)$

$(2,3)\ \hat{e}(T) = -\hat{e}(T) - (1,3)\hat{e}(T)$

$(1,2,3)\ \hat{e}(T) = (1,3)\hat{e}(T)$

$(1,3,2)\ \hat{e}(T) = -\hat{e}(T) - (1,3)\hat{e}(T)$

因此 R_{S_3} 在 (T) 的所有基是 在(T)，$(1,3)$在(T)。

以它的为基，我们可以求出 S_3 的表示矩阵

e.g. $(1,3,2)$

$$(1,3,2)\text{在}(T) = -\text{在}(T) - (1,3)\text{在}(T)$$

$$(1,3,2)(1,3)\text{在}(T) = (1,2)(1,3)(1,3)\text{在}(T)$$

$$= (1,2)\text{在}(T) = \text{在}(T)$$

所以表示矩阵为

$$\begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$$

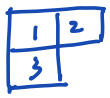
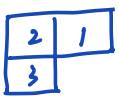
（与正规表示的思想类似，群元本身作基，群元本身作等名

$$\text{在}\hat{}(T)$$

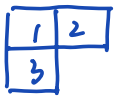
operator on basis

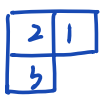
都是群元，其结果是 S_n 群的群乘规则下的乘法表对应。

设为 S_3 的两个二维不变子空间：

一个不变子空间：  

P $=$

 ：$(I-P_{13})(I+P_{12}) = I - P_{13} + P_{12} - P_{13}P_{12}$

 ：$(I-P_{23})(I+P_{21}) = I - P_{23} + P_{21} - P_{23}P_{21}$

P_{12} $= P_{12}(I-P_{13})(I+P_{12})$

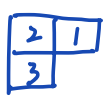
$= (P_{12} - P_{12}P_{13})(I+P_{12})$

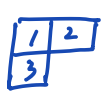
$= P_{12} - P_{12}P_{13} + I - P_{23}$

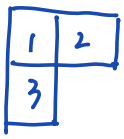
$P_{12} = P_{21}$

$P_{13}P_{21} = (2\ 1\ 3)$

$P_{12}P_{13} = (1\ 3\ 2)$ //

$\therefore$ P_{12} $=$ 

P_{12} $= (P_{12})^2$ $=$ 

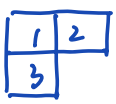
$\hat{P}_{23}$ 

$= \hat{P}_{23}(I - \hat{P}_{13})(I + \hat{P}_{12})$

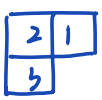
$= (\hat{P}_{23} - \hat{P}_{23}\hat{P}_{13})(I + \hat{P}_{12})$

$= \hat{P}_{23} - \hat{P}_{23}\hat{P}_{13} + \hat{P}_{23}\hat{P}_{12} - \hat{P}_{23}\hat{P}_{13}\hat{P}_{12}$

$= \hat{P}_{23} - \hat{P}_{23}\hat{P}_{13} + \hat{P}_{23}\hat{P}_{12} - \hat{P}_{13}$



: $(I - \hat{P}_{13})(I + \hat{P}_{12}) = I - \hat{P}_{13} + \hat{P}_{12} - \hat{P}_{13}\hat{P}_{12}$



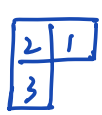
: $(I - \hat{P}_{23})(I + \hat{P}_{21}) = I - \hat{P}_{23} + \hat{P}_{21} - \hat{P}_{23}\hat{P}_{21}$

$\hat{P}_{23}\hat{P}_{13} = \hat{P}_{32}\hat{P}_{31} = (3\ 1\ 2)$

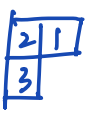
$\hat{P}_{13}\hat{P}_{12} = (1\ 2\ 3)$ //

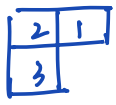
$\hat{P}_{23}$

$\stackrel{?}{=} a$

$+ b$ 

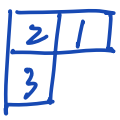
$=$

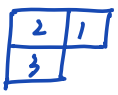
$-$ 

σ_{13}  $= \sigma_{13}(I - \sigma_{13})(I + \sigma_{12})$

$= (\sigma_{13} - I)(I + \sigma_{12})$

$= \sigma_{13} - I + \sigma_{23}\sigma_{12} - \sigma_{12}$

$= -$ 

$\sigma_{23}(\lambda$ $+ \mu$  $)$

$= \lambda($ $-$ $)$

$+ \mu($ $-$ $)$

$= \lambda$ $-(\lambda+\mu)$

$\hat{P}_{13}$

$= \hat{P}_{13}(I - \hat{P}_{13})(I + \hat{P}_{12})$

$= (\hat{P}_{13} - I)(I + \hat{P}_{12})$

$= \hat{P}_{13} - I + \hat{P}_{13}\hat{P}_{12} - \hat{P}_{12}$

$= -$

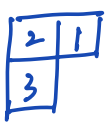
$\hat{P}_{13}$

$= \hat{P}_{13}(I - \hat{P}_{23})(I + \hat{P}_{12})$

$= \hat{P}_{13} - \hat{P}_{13}\hat{P}_{23} + \hat{P}_{13}\hat{P}_{12} - \hat{P}_{13}\hat{P}_{23}\hat{P}_{12}$

$= \hat{P}_{13} - \hat{P}_{13}\hat{P}_{23} + \hat{P}_{13}\hat{P}_{12} - \hat{P}_{2}$

$= -$

$+$ 

S_3 群只有3种对换 σ_{12}, σ_{23}, σ_{13}, 由3种对换的乘法可以得到所有的置换, 所以,

λ + μ 将构成不变子空间

求下表示矩阵

σ_{12} =

σ_{12} : $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ σ_{23} : $\begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}$

σ_{13} : $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}$ E : $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

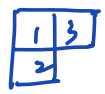
$\sigma_{12}\sigma_{13} = (1\,3\,2) = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$

$\sigma_{12}\sigma_{23} = \sigma_{21}\sigma_{23} = (2\,3\,1) = (1\,2\,3)$

$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix}$

A_3 : $\det = 1$, 其余的时为 -1

even perm odd perm

λ'  + μ'

将贡献相同的特征标

$\chi(\text{odd perm}) = 0$

$\chi(E) = 2$

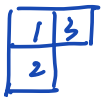
$\chi(D, F) = -1$

两个不相关的两个二维基空间提供了同一个不可约表示，贡献相同的特征标。

以 6 维基空间看 S_3

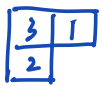
二维的不可约表示出现两次

但两次是不同的基空间

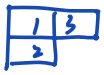
check λ'  + μ'

: $(I - P_{12})(I + P_{13})$

$= I - P_{12} + P_{13} - P_{12}P_{13}$

 : $(I - P_{23})(I + P_{31})$

$= I - P_{23} + P_{31} - P_{23}P_{31}$

P_{12}  $= P_{12}(I - P_{12})(I + P_{13})$

$= (P_{12} - I)(I + P_{13})$

$= P_{12} - I + P_{12}P_{13} - P_{13}$

$= -$

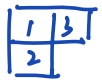
P_{12} $= P_{12}(I - P_{23})(I + P_{13})$

$= (P_{12} - P_{12}P_{23})(I + P_{13})$

$= P_{12} - P_{12}P_{23} + P_{12}P_{13} - P_{12}P_{23}P_{13}$

$= P_{12} - P_{12}P_{23} + P_{12}P_{13} - P_{23}$

$= -$ $+$

$\hat{P}_{23}$  $= \hat{P}_{23}(I - \hat{P}_{12})(I + \hat{P}_{13})$

$= (\hat{P}_{23} - \hat{P}_{23}\hat{P}_{12})(I + \hat{P}_{13})$

$= \hat{P}_{23} - \hat{P}_{23}\hat{P}_{12} + \hat{P}_{23}\hat{P}_{13} - \hat{P}_{23}\hat{P}_{12}\hat{P}_{13}$

$= \hat{P}_{23} - \hat{P}_{23}\hat{P}_{12} + \hat{P}_{23}\hat{P}_{13} - \hat{P}_{12}$

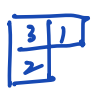
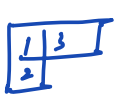
$=$ $-$

$\hat{P}_{23}$ $= \hat{P}_{23}(I - \hat{P}_{23} + \hat{P}_{31} - \hat{P}_{23}\hat{P}_{31})$

$= (\hat{P}_{23} - I + \hat{P}_{23}\hat{P}_{31} - \hat{P}_{23}\hat{P}_{23}\hat{P}_{31})$

$= \hat{P}_{23} - I + \hat{P}_{23}\hat{P}_{31} - \hat{P}_{31}$

$= -$

$\hat{P}_{13}$ $=$  $\hat{P}_{13}$ $=$ 

表示矩阵

$\hat{P}_{12}$ $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}$ $\hat{P}_{23}$: $\begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}$

$\hat{P}_{13}$ $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\hat{E}$: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\hat{P}_{12}\hat{P}_{23}$: $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$

$= (2\ 3\ 1)$

$\hat{P}_{12}\hat{P}_{13}$: $\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$

$= (1\ 3\ 2)$

χ : 三个对换为 0 $\rightarrow$ $\chi_{odd} = 0$

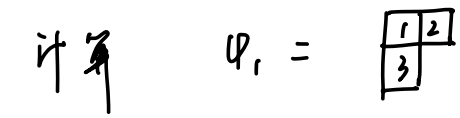
$\chi_E = 2$ $\chi_{even} = -1$

不同的基函数，相同的特征标。

（特征表示，但基之间不能用 S_3 连接）

From 王敏博： (App. 5.4)

$A_{12} S_{12} = 0$

计算 $\psi_1 =$  $= A_{13} S_{12}$

$\psi_2 =$ $= A_{23} S_{12}$

$(12)\psi_1 = \frac{1}{2}(S_{12} - A_{12}) A_{13} S_{12}$

$= \frac{1}{2} \underbrace{S_{12} A_{13} S_{12}}_{\| } - \frac{1}{2} \underbrace{A_{12} A_{13} S_{12}}_{\| }$

$(A_{13} + A_{23}) S_{12}$ $\quad (A_{13} - A_{23}) S_{12}$

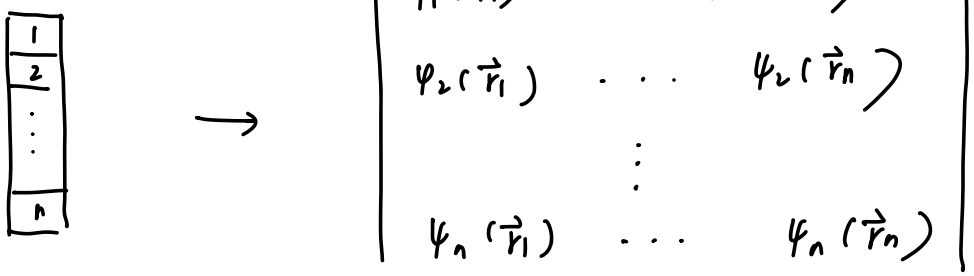
在此标记处理上下，可以写见

$(12)\psi_1 = a\psi_1 + b\psi_2$

$A_{13} S_{12}$ $\quad A_{23} S_{12}$

$S_{12} A_{12} = 0$ ， $S_{12} A_{22}$ 不对之

非零项： $S_{12} A_{ii}$ $\quad i \neq 1, 2$



$$= \sum_{\sigma} \varepsilon^{\sigma(1,2,\cdots n)} \psi_1(\vec{r}_{\sigma(1)})\,\psi_2(\vec{r}_{\sigma(2)}) \cdots \psi_n(\vec{r}_{\sigma(n)})$$

$\varepsilon^{\sigma(1,2,\cdots n)} = \mathrm{sgn}(\sigma)$

Hartree - Fock

n 阶 含反称张量

自动满足 Pauli 不相容原理

和交换反对称。

$$\Psi = \sum_{\sigma} \psi_1(\vec{r}_{\sigma(1)})\,\psi_2(\vec{r}_{\sigma(2)}) \cdots \psi_n(\vec{r}_{\sigma(n)})$$

含对称。

Symmetrization

Hartree state : product state

cyclic permutation: (轮换)

$(1\ 2\ 3\ \cdots\ n) = (2\ 3\ \cdots\ n\ 1)$

$= (3\ \cdots\ n\ 1\ 2)$

$\cdots$

$= (n\ 1\ 2\ \cdots\ 3)$

实现置换:

$1 \to 2,\ 2 \to 3,\ 3 \to 4,\ \cdots\ n \to 1$

$(1\ 2\ 3\ \cdots\ n) = (1\ n)\cdots(1\ 4)(1\ 3)(1\ 2)$

$1 \to 2$

$2 \to 1 \to 3$

$3 \to 1 \to 4$

$4 \to 1 \to 5$

$\cdots$

$n-1 \to 1 \to n$

$n \to 1$

A_3 $(g_1 + wg_2 + w^2 g_3) \sim (g_1 + wg_2 + w^2 g_3)$

P_{ij} $(g_1 + wg_2 + w^2 g_3) \sim (g_1 + w^2 g_2 + wg_3)$

$\begin{cases} g_1 + wg_2 + w^2 g_3 \quad (\varphi_1) \\ g_1 + w^2 g_2 + wg_3 \quad (\varphi_2) \end{cases}$

两种不变型

$P_3 \varphi_1 = w\varphi_1$

(123)

$P_3^2 \varphi_1 = w^2 \varphi_1$

$P_3^3 \varphi_1 = \varphi_1$

$P_n \longrightarrow P_1^S + P_{n-1}'$

$P_1^S = g_1 + g_2 + \cdots + g_n$

$P_3 \varphi_2 = g_2 + w^2 g_3 + wg_1 = w\varphi_2$

$P_{n-1}' =$

$\begin{pmatrix} \sum_{i=1}^{n} w^{i-1} g_i \\ \sum_{i=1}^{n} w^{2(i-1)} g_i \\ \vdots \\ \sum_{i=1}^{n} w^{(n-1)(i-1)} g_i \end{pmatrix}$ ← $n-1$ entries

$(1)(23)\varphi_1 = \varphi_2$

Circulant matrix

Gn n个事物为相对完备基

$n = 1 \oplus n-1$

$P_1^S \quad P_{n-1}'$

P_{n-1}' is an irreducible representation, which is present is every permutation group $P(n)$ (S_n)

$$\begin{array}{cccccc} \text{col}(1) & \text{col}(2) & \text{col}(3) & \cdots & \text{col}(n) & \\ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ w \\ \vdots \\ w^{n-1} \end{pmatrix} & \begin{pmatrix} 1 \\ w^2 \\ \vdots \\ w^{2(n-1)} \end{pmatrix} & \cdots & \begin{pmatrix} 1 \\ w^{n-1} \\ \vdots \\ w^{(n-1)(n-1)} \end{pmatrix} & \end{array} \qquad \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{pmatrix}$$

Fourier matrix

意味着 $n=p$ 时，

F_{n-1}' 是 g_4 是放下的 3 种不变量

$$P_n \begin{pmatrix} 1 \\ w \\ \vdots \\ w^{n-1} \end{pmatrix} = w \begin{pmatrix} 1 \\ w \\ \vdots \\ w^{n-1} \end{pmatrix}$$

$$P_n \begin{pmatrix} 1 \\ w^2 \\ \vdots \\ w^{2(n-1)} \end{pmatrix} = w^2 \begin{pmatrix} 1 \\ w^2 \\ \vdots \\ w^{2(n-1)} \end{pmatrix}$$

$$\vdots$$

$$(g_1 \quad g_2 \quad \cdots \quad g_n)\, p_n \begin{pmatrix} 1 \\ w \\ \vdots \\ w^{n-1} \end{pmatrix}$$

$$= w \left(\sum_{i=1}^{n} w^{i-1} g_i \right)$$

$$(g_1 \quad g_2 \quad \cdots \quad g_n)\, p_n \begin{pmatrix} 1 \\ w^2 \\ \vdots \\ w^{2(n-1)} \end{pmatrix}$$

$$= w^2 \left(\sum_{i=1}^{n} w^{2(i-1)} g_i \right)$$

$$(g_1 \quad g_2 \quad \cdots \quad g_n) \cdot \overset{col(i)}{\begin{pmatrix} \\ \\ \\ \end{pmatrix}}$$

$$p_n \, col(i) = w^{i-1} col(i)$$

Q: What is the key point?

本征矢是"旋不动"的矢量，与字母积为cyclic下的"不变理"

$$F_n = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}_{n \times n}$$

$$(F_n)_{ij} = w^{ij}$$

$$i, j = 0, 1, 2, \cdots, n-1$$

$$w^n = 1$$

$$w = e^{i\pi/n}$$

$$F_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{pmatrix}$$

remarkable matrix

Circulant matrix : Each row is a cyclic shift of the previous row

$$C = \begin{pmatrix} c_0 & c_1 & c_2 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & c_1 & \cdots & c_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & c_3 & \cdots & c_0 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

和 4 个本征矢排成列向量

$$P^4 = I$$

P 的本征值为 $1, i, i^2, i^3$

$$P \begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{pmatrix} = \begin{pmatrix} g_2 \\ g_3 \\ g_4 \\ g_1 \end{pmatrix}$$

$P = (1\ 2\ 3\ 4)$ cyclic

$(1)\quad (2\ 3\ \varphi)$

$$\begin{pmatrix} 1 & & & \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_\varphi \end{pmatrix} = \begin{pmatrix} g_1 \\ g_3 \\ g_\varphi \\ g_2 \end{pmatrix}$$

基组 $P(3)$ 的二维表示构造：

$\{\sin\theta, \cos\theta\}$

$\{g_1 + \omega g_2 + \omega^2 g_3,\ g_1 + \omega^2 g_2 + \omega g_3\}$

$\{$, $\}$ or

$\{$, $\}$

Tray method $= P_1^S + P_2$

$1 \oplus 2$

但表示矩阵并非直接分块